

CLASS 9th

PHYSICS BASIC NOTES

A. Motion

- Distance vs Displacement: Distance = total path, Displacement = straight line from start to end
- Speed & Velocity:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}, \text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

- Acceleration: Rate of change of velocity

$$a = \frac{v - u}{t}$$

- Equations of Motion (for uniformly accelerated motion):

1. $v = u + at$

2. $s = ut + \frac{1}{2}at^2$

3. $v^2 = u^2 + 2as$

B. Force & Laws of Motion

- Newton's Laws:
 - Law of Inertia
 - $F = ma$
 - Action-Reaction
- Momentum: $p = mv$
- Impulse: Change in momentum

C. Gravitation

- Universal Law of Gravitation:

$$F = G \frac{m_1 m_2}{r^2}$$

- Acceleration due to gravity: $g = 9.8 \text{ m/s}^2$
- Weight: $W = mg$

D. Work, Energy & Power

- Work: $W = F \cdot d \cdot \cos \theta$
- Kinetic Energy (KE): $KE = \frac{1}{2}mv^2$
- Potential Energy (PE): $PE = mgh$
- Power: $P = \frac{W}{t}$
- Law of Conservation of Energy: Total energy remains constant

E. Sound

- Speed of Sound depends on medium
- Frequency (f) and Wavelength (λ): $v = f\lambda$
- Echo & Reverberation

F. Light

- Reflection & Refraction: Laws of reflection, Snell's Law
- Lenses & Mirrors: Focal length, image formation
- Dispersion of Light: Splitting into spectrum

G. Gravitation & Pressure

- Pressure: $P = \frac{F}{A}$
- Archimedes Principle: Upthrust = weight of displaced fluid